

THEORY AND METHODS
PROBABILISTIC INVERSION FORMULAS FOR
THE LAPLACE TRANSFORM

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Summary: We give probabilistic proofs of a number of real inversion formulas for the one-sided Laplace transform. For many of the quoted results, simplified proofs are given. Also a number of new inversion formulas are included.

1. The General Approach

Assume that μ is a bounded measure on the positive half line. We define its Laplace transform by

$$g(s) := \int_0^\infty e^{-sx} d\mu(x)$$

where $s \geq 0$.

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Assume that $d_n(u)$ is a sequence of bounded functions on the positive halfline for which as $n \rightarrow \infty$

$$d_n(u) \rightarrow \begin{cases} 0 & \text{if } u < 1, \\ 1/2 & \text{if } u = 1, \\ 1 & \text{if } u > 1. \end{cases} \quad (1)$$

Then a *potential* inversion formula is for $0 \leq y_1 < y_2$

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} \int_0^\infty \left\{ d_n\left(\frac{u}{y_1}\right) - d_n\left(\frac{u}{y_2}\right) \right\} d\mu(u) \quad (2)$$

where the left-hand side is an abbreviation for

$$\mu\{y_1; y_2\} := \frac{1}{2}\mu\{y_1\} + \mu(y_1, y_2) + \frac{1}{2}\mu\{y_2\}.$$

In turn $\mu\{y\}$ stands for the weight or measure at the point y while $\mu(a, b)$ is shorthand for the measure of the open interval (a, b) .

If μ has a derivative $\mu'(y) := \frac{d\mu(y)}{dy}$ then the above formula should be adapted to immediately give an expression for μ' . Put $y_1 = y$ and $y_2 = y + \epsilon$ in (2) and apply l'Hopital's rule to find that such a formula is

$$\mu'(y) = \frac{d\mu(y)}{dy} = \lim_{n \rightarrow \infty} \int_0^\infty \frac{u}{y^2} d'_n\left(\frac{u}{y}\right) d\mu(u) \quad (3)$$

where we assumed that $d_n(u)$ has a derivative for all values of $u \neq 1$.

The sequences $\{d_n(u)\}$ will be constructed by using probabilistic methods. The required asymptotics will follow by appealing to classical results from probability theory.

Inversion formulas of the type (2) can be important if the Laplace transform is only known on the positive half line. Moreover, there is some virtue in providing real variable techniques in an area where traditionally complex integration is used for the inversion of the Laplace transform. A further benefit is that some of the formulas can give approximations when infinite sums are replaced by their finite counterparts.

2. The Post-Widder Formula

This is one of the most classical inversion formulas and it can be found in Widder (1941). We generalized the latter approach in Teugels (1990) following an innovative probabilistic procedure for the derivative formula by Feller in Feller (1971). We nevertheless include the result because of its simplicity and since we like to return to it in the final section.

Assume that $\{X_i, i = 1, 2, \dots\}$ is a sequence of iid rv with common exponential distribution with mean 1 and let $S_n = X_1 + X_2 + \dots + X_n$. Take $d_n(u) = P\{S_n \leq nu\}$. Then the law of large numbers implies that $d_n(u)$ converges to 0 when $u < 1$, while $d_n(u) \rightarrow 1$ when $u > 1$. For $u = 1$, write $d_n(1) = P\{S_n - n \leq 0\} \rightarrow \Phi(0) = 1/2$ in view of the central limit theorem. Therefore, the sequence $\{d_n(\cdot)\}$ satisfies the requirements of (1).

Since the sum of independent exponentials with mean 1 follows an Erlang distribution, we have the explicit expression

$$d_n(u) = \frac{1}{n!} \int_0^{nu} e^{-v} v^n dv.$$

The inversion formula readily follows in that

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} \int_{y_1}^{y_2} \frac{(-n)^n}{\Gamma(n)} g^{(n)}\left(\frac{n}{t}\right) \frac{dt}{t^{n+1}}$$

where $g^{(n)}(\cdot)$ denotes the n -th derivative of the Laplace transform g .

The derivative version is

$$\mu'(y) = \lim_{n \rightarrow \infty} \frac{n^{n+1}}{n!} \frac{(-1)^n}{y^{n+1}} g^{(n)}\left(\frac{n}{y}\right).$$

3. The Widder Formula

Also this formula can be found in Widder (1941). A probabilistic proof can be based on a well-known property of the Poisson distribution. Take

$$d_n(u) = \sum_{m=0}^{[n]} e^{-nu} \frac{(nu)^m}{m!}.$$

Then the weak law and the central limit theorem again show that this sequence satisfies the necessary requirements. Hence one quickly arrives at the inversion formula

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} \sum_{m=[ny_1]+1}^{[ny_2]} \frac{(-n)^m}{m!} g^{(m)}(n).$$

The Widder formula seems particularly attractive. Define

$$F_n(t) := \sum_{m=0}^{[nt]} \frac{(-n)^m}{m!} g^{(m)}(n).$$

Then its Laplace transform turns out to be

$$\int_0^\infty e^{-st} dF_n(t) = g(n - ne^{-s/n}).$$

When $n \rightarrow \infty$ then the right-hand side tends to $g(s)$. So there exists a limiting function F such that $F_n \Rightarrow F$ which gives a short proof of the famous Bernstein theorem on completely monotone functions, see Widder (1941). Note that the quantity $x(1 - e^{-s/x})$ is monotonically increasing in x . Hence the convergence of the transforms of F_n to g is from below. This implies *inequalities* in the Laplace ordering relation discussed in Denuit et al. (2005).

4. The Shohat-Tamarkin Formula

After treating the original formula, we give a slight extension.

4.1 Original Formula

In Shohat and Tamarkin (1943), a remarkable inversion formula is given as the solution of a moment problem on the positive half line. In this section we give a probabilistic proof of this important result.

Assume that $\{X_i, 1 \leq i\}$ is a sequence as in (2) above. Let further Y_n be an independent Poisson counting variable with mean n , i.e.

$$P\{Y_n = k\} = e^{-n} \frac{n^k}{k!}.$$

Look at the distribution of the random sum S_{T_n} where $T_n = Y_n + 1$. If we write

$$\frac{S_{T_n} - n}{\sqrt{n}} = \sqrt{\frac{T_n}{n}} \frac{S_{T_n} - T_n}{\sqrt{T_n}} + \frac{T_n - n}{\sqrt{n}}$$

then a Slutsky-type argument shows that the quantity on the left converges in distribution to a normal variable with mean zero. Hence $d_n(u) = P\{S_{T_n} \leq nu\}$ satisfies the necessary requirements (1).

We now rewrite $d_n(u)$ in order to relate it to the Laplace transform. By a conditioning argument one can write

$$d_n(u) = \sum_{k=0}^{\infty} P\{Y_n = k\} P\left\{\sum_{i=1}^{k+1} X_i \leq u\right\} = \sum_{k=0}^{\infty} e^{-n} \frac{n^k}{k!} \frac{1}{\Gamma(k+1)} \int_0^{nu} e^{-v} v^k dv.$$

A short calculation shows that

$$d_n(u) = n \int_0^u e^{-(1+x)n} I_0(2n\sqrt{x}) dx$$

where $I_0(\cdot)$ is the modified Bessel function of order zero.

Recall the *Hardy-Hille formula*

$$\frac{1}{1-z} \exp\left(-\frac{(x+y)z}{1-z}\right) I_0\left(\frac{2\sqrt{xyz}}{1-z}\right) \quad (4)$$

that links the modified Bessel function $I_0(\cdot)$ to the Laguerre polynomials

$$L_m(y) = \sum_{k=0}^m \binom{m}{k} \frac{(-y)^k}{k!}. \quad (5)$$

In (4) we replace z by $1 - \frac{u}{n}$ and y by $\frac{ux}{z}$. Plugging in the resulting expression for the function $I_0(\cdot)$ in the potential inversion formula (2)

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} n \int_0^\infty \int_{y_1/u}^{y_2/u} e^{-n(1+x)} I_0(2n\sqrt{x}) d\mu(u) dx$$

quickly leads to the expression

$$\mu\{y_1; y_2\} = \int_{y_1}^{y_2} \sum_{m=0}^{\infty} L_m(v) \int_0^\infty e^{-u} L_m(u) d\mu(u) dv.$$

If we now use the explicit form of the Laguerre polynomial then we arrive at the Shohat-Tamarkin formula

$$\mu\{y_1; y_2\} = \int_{y_1}^{y_2} \sum_{m=0}^{\infty} \sum_{k=0}^m \frac{m!}{k!^2 (m-k)!} g^{(k)}(1) L_m(u) du.$$

After an interchange of summation, a derivative version is

$$\mu'(y) = \sum_{k=0}^{\infty} \frac{g^{(k)}(1)}{k!^2} \sum_{m=k}^{\infty} \frac{m!}{(m-k)!} L_m(y). \quad (6)$$

Note that the second summation does not depend on g .

4.2 Generalization

Following the same pattern as in the previous section, we can slightly generalize the Shohat-Tamarkin approach in that we can use the Laplace transform and its derivatives in an arbitrary point on the positive half line. To do that, start from

$$d_n(u) = na \int_0^u e^{-n(1+ax)} (\sqrt{ax})^b I_b(2n\sqrt{ax}) dx$$

where I_b is the modified Bessel function of order b and a is any positive number.

The general form of the Hardy-Hille formula links the Bessel function I_b to the generalized Laguerre polynomials in that

$$\sum_{n=0}^{\infty} \frac{n!}{\Gamma(n+b+1)} L_n^b(x) L_n^b(y) z^n = \frac{(xyz)^{-b/2}}{1-z} \exp\left(-\frac{(x+y)z}{1-z}\right) I_b\left(\frac{2\sqrt{xyz}}{1-z}\right).$$

Then using almost identical operations as before we find

$$\mu\{y_1; y_2\} = \int_{y_1}^{y_2} \sum_{n=0}^{\infty} \lambda_n(a) a^{b+1} \frac{\Gamma(n+1)}{\Gamma(n+b+1)} u^b L_n^b(au) du$$

where in turn

$$\lambda_n(a) = \sum_{k=0}^n \binom{n+b}{n-k} \frac{a^k}{k!} g^{(k)}(a).$$

The choice $a = 1$ and $b = 0$ leads back to the Shohat-Tamarkin formula.

The derivative formula reads

$$\mu'(y) = \sum_{k=0}^{\infty} \frac{a^{k+b+1} g^{(k)}(a)}{k! \Gamma(b+k+1)} \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} y^b L_n^b(ay).$$

5. Gaver-Stehfest Formula

We now turn to a formula derived in Gaver (1966) and Stehfest (1970). Assume that $\{X_i, i = 1, 2, \dots\}$ is again a sequence of iid rv with common exponential distribution with mean 1. Independent of this, let $\{Y_i, i = 1, 2, \dots\}$ be another such sequence. It is a simple exercise to prove that

$$P\left\{\frac{\sum_{i=1}^{n+1} Y_i}{\sum_{j=1}^n X_j} \leq y\right\} = n \binom{2n}{n} \int_{(1+y)^{-1}}^1 t^{n-1} (1-t)^n dt,$$

a slight modification of a beta-distribution. We use this to produce another useful sequence $d_n(\cdot)$.

Define

$$d_n(u) = P\left\{-\frac{1}{\log 2} \log \left\{\frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i + \sum_{j=1}^{n+1} Y_j}\right\} \leq u\right\}.$$

It then readily follows that

$$d_n(u) = n \binom{2n}{n} \int_{2^{-u}}^1 t^{n-1} (1-t)^n dt.$$

That the sequence $d_n(u)$ satisfies the limiting requirements easily follows from the law of large numbers and the central limit theorem. The potential inversion formula therefore reads

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} n \binom{2n}{n} \int_0^\infty \int_{2^{-u/y_1}}^{2^{-u/y_2}} t^{n-1} (1-t)^n dt d\mu(u).$$

Expansion of the factor $(1-t)^n$ followed by the integration with respect to t quickly leads to the Gaver-Stehfest formula

$$\mu\{y_1; y_2\} = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k n}{n+k} \binom{2n}{n} \binom{n}{k} \left[g\left(\frac{n+k}{y_2} \log 2\right) - g\left(\frac{n+k}{y_1} \log 2\right) \right].$$

Again a slight generalization is possible when we can replace $\log 2$ by $\log(1+a)$ for any $a > 0$ while adapting the coefficients. If μ comes from a probability measure without mass at the origin, then its Laplace transform takes on the values $0 = g(\infty)$ and $1 = g(0)$. Using either $y_1 = 0$ or $y_2 = \infty$ simplifies the formula somewhat.

6. Distributions

If μ is a probability measure linked to a distribution F , then we can find more formulas by using the *integrated tail* of F defined by the expression

$$F_1(y) := \int_0^y (1 - F(x)) dx.$$

The Laplace transform of this measure, say $g_1(s)$, is given by $g_1(s) = s^{-1}(1 - g(s))$. We can now use the derivative version of any inversion formula to obtain an expression for F .

6.1 The Shohat-Tamarkin Formula

We illustrate the procedure with the Shohat-Tamarkin formula. Then (5) implies

$$1 - F(y) = \sum_{k=0}^{\infty} \frac{g_1^{(k)}(1)}{k!^2} \sum_{m=k}^{\infty} \frac{m!}{(m-k)!} L_m(y) =: \sum_{k=0}^{\infty} \frac{g_1^{(k)}(1)}{k!^2} c_k(y).$$

However, for $k \geq 1$,

$$D^k g_1(s) = \frac{d^k}{ds^k} \frac{1 - \hat{F}(s)}{s} = \frac{(-1)^k k!}{s^{k+1}} (1 - g(s)) - \sum_{\ell=1}^k \frac{(-1)^{k-\ell} k!}{\ell! s^{k-\ell+1}} g^{(\ell)}(s). \quad (7)$$

Using (7) at $s = 1$, an easy calculation leads to

$$1 - F(y) = (1 - g(1)) \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} c_k(y) - \sum_{\ell=1}^{\infty} \frac{g^{(\ell)}(1)}{\ell!} \sum_{k=\ell}^{\infty} \frac{(-1)^{k-\ell}}{k!} c_k(y).$$

Inserting the expression for the quantities $c_k(\cdot)$ we find that

$$1 - F(y) = (1 - g(1)) L_0(y) - \sum_{\ell=1}^{\infty} \frac{g^{(\ell)}(1)}{\ell!} \sum_{n=\ell}^{\infty} \alpha_{n,\ell} L_n(y) \quad (8)$$

where

$$\alpha_{n,\ell} := \sum_{k=\ell}^n (-1)^{k-\ell} \binom{n}{\ell}.$$

The first term in (8) can be incorporated since $\alpha_{n,0} = \delta_{n,0}$. For the distribution itself we have

$$F(y) = \sum_{\ell=0}^{\infty} \frac{g^{(\ell)}(1)}{\ell!} \sum_{n=\ell}^{\infty} \alpha_{n,\ell} L_n(y).$$

6.2 The Post-Widder Formula

As a second illustration, take the Post-Widder formula. If we use (7) in

$$1 - F(y) = \lim_{n \rightarrow \infty} \frac{n^{n+1}}{n!} \frac{(-1)^n}{y^{n+1}} g_1^{(n)}(s) \Big|_{s=n/y}$$

then the inversion formula simplifies to the expression

$$F(y) = \lim_{n \rightarrow \infty} \sum_{\ell=0}^n \frac{1}{\ell!} \left(-\frac{n}{y} \right)^{\ell} g^{(\ell)} \left(\frac{n}{y} \right).$$

The latter formula has been derived by Stadtmüller and Trautner (1981).

7. Conclusions

The variety of potential inversion formulas is of course enormous and one needs some guidelines which of them is better for any practical situation. Let us just point out the following:

- In the Post-Widder formula, all derivatives of g are used on the entire positive half line.
- The Widder formula uses g on an infinite lattice.
- The Shohat-Tamarkin result employs the derivatives of g in a single point.
- The Gaver-Stehfest uses only g but on the entire positive half line.

Of course, since $g(s)$ is an analytic function as long as $\Re s \geq 0$, all of these expressions are internally linked by analytic continuation.

We should add that the inversion problem becomes even broader if we allow μ to be spread over the entire real line. All formulas resulting from such an approach are also applicable to the case of the non-negative half line. We hope to return to this issue in a later contribution.

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